CV

LUCAS GOLOBOVANTE

SUMMARY

Computer Engineering graduate (Bachelor's Degree) specializing in Data Science and Artificial Intelligence, transitioning to an AI Engineer Trainee at Siemens S.A. following a Machine Learning curricular internship. I have practical experience developing Deep Learning and Computer Vision solutions, including a production Machine Learning pipeline, along with solid knowledge in data analysis using Power BI and SQL.

I stand out for my ability to work both in teams and individually, being organized and motivated by new challenges and continuous learning.

PROFESSIONAL EXPERIENCE

AI Engineer Trainee

Siemens S.A. (Smart Infrastructure)

Starting 09/2026 | Porto, Portugal

- Continuing full-time within the Smart Infrastructure team following the curricular internship below, applying Machine Learning and Computer Vision to engineering documentation workflows.

Curricular Internship - Machine Learning for Smart Infrastructure Service Software

Siemens S.A. (Smart Infrastructure)

03/2026 - 08/2026 | Porto, Portugal

- Designed and implemented a two-stage Computer Vision and Deep Learning pipeline to detect and classify engineering symbols in Air Handling Unit technical drawings, combining a classical CV preprocessing and region-proposal stage (OpenCV: binarization, morphological operations, Hough transform, connected-component analysis) with a MobileNetV2 classifier trained via transfer learning (TensorFlow/Keras) across 22 symbol classes, achieving approximately 97% accuracy on an independent held-out test set. Addressed class imbalance through class weighting and data augmentation, and developed a spatial co-occurrence analysis (DBSCAN clustering and association-rule mining) to uncover functional relationships between detected components. Deployed the pipeline to Microsoft Azure, with the inference job running on a VM via a scheduled job.yaml and results stored in Blob Storage, and model training also carried out on Azure, while collaborating in an Agile/Scrum cross-functional team (PM, RTE, developers) with daily stand-ups and weekly technical knowledge-sharing sessions.

EDUCATION

BSc in Computer Engineering

Polytechnic University of Coimbra
09/2023 - 07/2026 | Coimbra, Portugal

- Final Average: 14/20
- Completed the Information Systems branch, focusing on the complete data lifecycle: from ingestion and modeling in relational databases (Oracle/SQL) to transformation into intelligence via Machine Learning and Business Intelligence. Specialized in system architecture and business process analysis, developing scalable solutions integrating modern technologies like Python, C++, and data visualization frameworks. Final project: development of a Machine Learning and Computer Vision pipeline for Siemens S.A.

Secondary Education - Sciences and Technologies

Escola Secundária da Maia
09/2020 - 06/2023 | Maia, Porto

- Final grade: 17/20

Basic Education

Luanda International School
Colégio Angolano de Talatona - Cambridge

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C#, C++, SQL (Oracle/PLSQL), Java, MATLAB, Assembly
- **Data Science & AI:** Machine Learning (Pandas, Scikit-learn, NumPy, Seaborn, OpenCV), Deep Learning
- **Data Visualization & BI:** Power BI, Data warehousing
- **Web Development:** React.js, JavaScript, HTML, CSS, Blazor
- **Systems & Tools:** Linux, GitHub, VS Code, Microsoft Azure, DevOps/MLOps

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ACADEMIC PROJECTS

Machine Learning Model for Recycling

Developed a Computer Vision system based on Deep Learning for the automatic classification of recyclable waste. Utilized Transfer Learning (MobileNetV2) and hyperparameter optimization via Swarm Intelligence (GWO), featuring a Web interface for real-time usage. Achieved 95% final accuracy in testing.

VendingPLUS Logistics Management System

Designed the server architecture and business logic using Oracle SQL and PL/SQL. Implemented Stored Procedures, Triggers, and complex functions for stock automation, route optimization, and data integrity.

Web Data Integration System

Developed an ETL solution using Java and Regular Expressions to extract and unify data from heterogeneous sources. Implemented data validation with XML/XSD and dynamic report generation via XSLT and XQuery.

Autonomous Taxi Fleet Simulation

Simulated an autonomous taxi management platform developed in C for Linux/UNIX environments. Key Concepts: Named pipes, Anonymous pipes, Threads, Fork + Exec.

Business Intelligence System - Port of Figueira da Foz

Developed a Business Intelligence (BI) system for the Administration of the Port of Figueira da Foz. Transformed dispersed operational data into strategic information for decision-making (Flow: CSV/MySQL -> ETL -> SQL Server -> Power BI).

Garden Ecosystem Simulator (C++)

Developed a console-based ecosystem simulator focused on the practical application of Object-Oriented Programming in C++. Utilized inheritance, polymorphism, memory management and a command interpreter.

LANGUAGES

- Portuguese: Native
- English: Advanced (C1)

CONTACT

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